

Database Models

Repositories Collection: (MongoDB)

```
{
  repoURL: string (the repo URL for each ingested repo will be unique since no two repos can have the same URL. Main usage will be
to connect to its respective chunks and act as an identifier for each repo)
  latestSHA: string (the commit SHA as of ingestion will update when re-ingested and will act as a guard to check if a repo should be
re-ingested)
  lastAccessed: date (the last time this repo was used, it updates every time a user queries that repo, and it will be used to determine
which repo gets deleted when space is needed)
  fileTree: object (holds a file tree of the valid files of an ingested repo)
  timestamps: date
}
```

Chunks Collection: (MongoDB)

```
{
  content: string (holds the unaltered code block)
  embedding: [number] (holds the embedding of the code block)

  metadata: {
    repoURL: string (the repo URL of the associated repo)
    relativePath: string (the relative file path of where this repo is located)
    fileName: string (the name of the file this code block is in)
    name: string (the name of the code block, whether it be a class, function, or other)
    type: string (class, function, block, other)
    language: string (what the language of the code block is)
    parentDir: string (the parent dir name of this file)
    startLine: number (what line number the code block starts at)
    endLine: number (what line number the code block ends at)
  }

  timestamps: date
}
```

Semantic Cache:

```
{
  repoURL: string (the repo URL of the associated repo)
  query: string (the question that was saved)
  queryEmbedding: [number] (embedding of the query used for search)
  response: string (the response that was saved)
  contextStats: {} (stats like used files and chunks used count)
  timestamps: date
}
```

Ingest Progress:

```
{
  repoURL: string (the repo URL of the associated repo)
  status: "idle" | "processing" | "complete" | "error"
  statusStage: "complete" | "error" | "cloning" | "scanning" | "storageCheck" | "chunking" | "embeddingAndProcessing"
  statusMessage: string
  statusMeta: any
  statusHistory: any[]
  statusUpdatedAt: NativeDate
}
```